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# PPAPlace: Enhancing Chip Placement via Mixed-Size PPA Prediction

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## Summary

This paper proposes PPAPlace, a placer-agnostic framework that learns to predict post-route PPA (WNS/TNS/power/area) from mixed-size placement states using a dual-stream model (graph attention over the macro connectivity and CNN over a rasterized placement grid). A systematic study shows that labels from post-global-routing (GRT) correlate far better with final post-route timing than pre-route or post-CTS metrics, at roughly one-sixth the cost of full routing, and that combining post-GRT labels with mixed-size features materially improves ranking accuracy. Using the predictor to score candidate placements (demonstrated over DREAMPlace hyperparameter configurations), PPAPlace selects placements that improve post-route TNS by 33% and WNS by 18% over Hier-RTLMP and outperform LaMPlace on five ChiPBench circuits.

## Strengths

**Technical novelty and innovation**

- Introduces a principled, dual-stream predictor that jointly captures spatial density/congestion patterns (RUDY, pin density, macro occupancy) and macro-level connectivity via GAT, going beyond macro-only surrogates.
- Provides a careful label-fidelity analysis across design-flow stages, identifying post-GRT as a robust and cost-effective supervision point for cross-stage learning.
- Proposes a placer-agnostic scoring formulation that can, in principle, guide configuration search for arbitrary placement engines without altering their internals.

**Experimental rigor and validation**

- Conducts a cross-stage correlation study across 10 diverse ChiPBench circuits; ablations disentangle contributions of label stage and representation (2×2 factorial) and compare predictor variants (macro-only, CNN-only, GAT-only, GAT+CNN).
- Evaluates end-to-end on five unseen designs through a complete RTL-to-signoff flow; includes LOCO cross-circuit validation and top-k selection utility.
- Reports runtime breakdown and a cost-matched labeling study (many post-GRT vs few post-DRT samples), offering practical guidance for data collection.

**Clarity of presentation**

- Clearly explains the architecture, features, and loss (MSE + pairwise ranking), along with data normalization and cross-circuit generalization strategies.
- Results are easy to interpret (Kendall's tau for ranking, normalized PPA metrics for end-to-end), and limitations are explicitly acknowledged.

**Significance of contributions**

- Targets a well-recognized HPWL-to-PPA mismatch and demonstrates measurable end-to-end timing gains over strong baselines.
- Offers a methodology (post-GRT labeling + mixed-size features) with implications for a broad class of cross-stage learning methods.

## Weaknesses

**Technical limitations or concerns**

- "Placer-agnostic" claim is only partially validated: all selection experiments are over DREAMPlace configurations; no experiments with hierarchical, RL/diffusion, or evolutionary macro placers despite



[WNS, TNS, Power, Area] differences without clarifying directionality/weighting (e.g., WNS sign conventions and trade-off handling), raising concerns about stability and objective alignment.

- Pairwise ranking over all in-circuit pairs can be quadratic in sample count; the paper does not specify sampling or efficiency strategies for scalability.

**Experimental gaps or methodological issues**

- Missing comparisons to timing-driven analytical baselines (e.g., DREAMPlace 4.0 with timing-driven net weighting, or recent timing-driven GPU placers), which are directly relevant and could reduce the reported performance gap.
- LaMPlace results are taken from the literature rather than re-run in the same pipeline; this weakens claims of superiority due to possible setup differences.
- The end-to-end results focus on WNS/TNS; power/area predictions are claimed but not thoroughly evaluated for downstream QoR improvements.
- The runtime table appears inconsistent (e.g., aggregate times for AutoDMP and the cost of K=500 placement configurations vs. a single full-flow run), and apples-to-apples runtime comparisons (counting all stages) are unclear.

**Clarity or presentation issues**

- The treatment of objective signs and normalization for multi-objective ranking and selection (argmin over a composite score) is under-specified.
- Some details of the grid features (e.g., how RUDY and net-bbox density are computed with obstacles/macros across layers) and edge construction nuances (hyperedges to cliques) are glossed over.

**Missing related work or comparisons**

- No comparison against recent timing-driven extensions to DREAMPlace (e.g., momentum-based net weighting or path-based

objectives) and advanced timing-driven placers that report strong TNS/WNS gains.

- Academic macro-placement frameworks that explicitly target PPA improvements (e.g., Re2MaP) are not discussed as baselines; this would strengthen context and claims.

## Detailed Comments

**Technical soundness evaluation**

- The rationale for post-GRT labeling is well supported by strong and consistent rank correlations across diverse designs, and the 2×2 ablation gives convincing evidence that both label fidelity and mixed-size representation matter.
- The dual-stream architecture is technically reasonable: the CNN captures congestion/density/routability cues, while GAT provides structural macro connectivity. Using GAP for size invariance is standard and appropriate.
- The ranking loss as a dot product over multi-metric deltas is elegant but under-specified: without explicit weighting or sign unification, improvements in one metric could be canceled by regressions in another; clarifying how this is resolved is important for robustness.
- Per-circuit z-score normalization encourages learning relative patterns and helps cross-design transfer; this is sensible in light of absolute PPA variations across circuits.

**Experimental evaluation assessment**

- The label-stage study is a highlight and appears carefully controlled, with consistent flows and within-circuit comparability. Reporting both Spearman and Kendall correlations across multiple circuits is welcome.
- The architecture ablation supports the claim that both spatial and graph streams contribute; tau=0.31 is moderate but practically useful for top-k selection.
- End-to-end results on five unseen designs through a full flow are strong and show clear improvements over Hier-RTLMP and better results than reported LaMPlace numbers. Still, relying on published

LaMPlace results is a limitation; re-running would strengthen the evidence.

- Power/area predictions are not validated as end-to-end improvements; including these would deliver on the claimed multi-metric prediction.
- Runtime analysis is difficult to interpret; generation of 500 candidates plus one full flow seems substantially more expensive than a single baseline run, yet the table suggests otherwise for some baselines. A fair, wall-clock, stage-by-stage accounting across methods under a matched budget would make the comparison more credible.

**Comparison with related work (using the summaries provided)**

- ChiPBench emphasizes the weak correlation between HPWL and timing; PPAPlace aligns directly with its call to “shift left” toward end-to-end QoR and demonstrates concrete gains.
- Compared to LaMPlace (macro-only, pre-route labels), PPAPlace’s mixed-size input and post-GRT labels are principled improvements. However, controlled re-evaluation is needed for a definitive comparison.
- Recent timing-driven placement (e.g., DREAMPlace 4.0 net weighting; Efficient timing-driven frameworks) should be included in baselines. Since they target timing more directly than raw HPWL, they are a closer competitor to PPA-aware selection.
- Re2MaP and BBO-based frameworks (BBOPlace-Bench) highlight that strong heuristic/BBO methods can improve PPA when objectives are aligned; discussing/including them would better position PPAPlace.
- MasterRTL and ParaGate operate at different pipeline stages (RTL and parasitics/timing reasoning), but both underscore the importance of stage-aware labels and domain transfer. PPAPlace’s post-GRT choice is consistent with this broader trend toward higher-fidelity supervision.

**Discussion of broader impact and significance**

- The work advances a practical recipe—use post-GRT supervision and mixed-size features—to close the HPWL-to-PPA gap. This has immediate applicability for design-space exploration and autotuning of placement flows.
- If the “placer-agnostic” promise is validated across more paradigms (hierarchical, RL/diffusion, BBO), this paradigm could become a standard component of PD toolchains to triage candidate solutions before costly routing.

- The release of code/data would help the community converge on better cross-stage predictors and provide a benchmark for future methods.

## Questions for Authors

1. How are the multi-objective differences (WNS, TNS, Power, Area) aligned and weighted in the ranking loss and in the final selection score? Are WNS/TNS signs inverted to ensure "lower is better," and are there per-metric weights?
2. Do you subsample pairs for the ranking loss to avoid O(N^2) growth within each circuit? If so, what sampling policy and ratio are used, and how sensitive are results to this choice?
3. Can you report end-to-end power and (if applicable) DRC metrics for the selected placements to support claims beyond timing?
4. How does PPAPlace compare against timing-driven DREAMPlace variants (e.g., DREAMPlace 4.0 momentum-based net weighting) or recent path-based timing-driven placers? Including these baselines would strengthen the case.
5. For "placer-agnostic" validation, can you provide experiments where the candidate pool comes from a different paradigm (e.g., Hier-RTLMP weight sweeps, EA/RL macro placers), and show that the predictor still selects better candidates?
6. The runtime table suggests surprisingly low aggregate times for AutoDMP and the K=500 candidate generation. Could you provide a strict apples-to-apples wall-clock breakdown (placement, CTS, GRT, DRT) per candidate and per method, under a fixed budget?
7. How sensitive is the predictor to the 64×64 grid resolution, and does increasing channels (e.g., per-layer RUDY, blockage maps) further improve accuracy?
8. What is the effect of using post-DRT labels for fine-tuning after training with post-GRT labels (e.g., a small fraction of DRT labels)? Is there a data-efficiency sweet spot?
9. In LOCO evaluation, tau varies across circuits. Can you analyze failure modes (e.g., macro-rich vs. macro-light, high I/O density, congested

topologies) and how features might be adapted?

10. Do you observe any overfitting to DREAMPlace-specific artifacts in features (e.g., density patterns)? How do predictions behave on placements produced by other tools?

## Overall Assessment

This paper addresses an important and timely problem—bridging the gap between wirelength-optimized placement and true post-route PPA—and offers a sound, empirically supported approach: use post-GRT supervision and mixed-size features to train a cross-stage predictor, and deploy it to select promising placements without running full routing on all candidates. The correlation study is careful and practically informative, and the end-to-end gains over a strong hierarchical baseline are compelling. That said, some claims would be stronger with broader validation: "placer-agnostic" utility is only shown for DREAMPlace configurations; comparisons omit timing-driven analytical baselines; and LaMPlace numbers are taken from prior work rather than re-run. The multi-objective ranking/selection formulation needs clearer specification, and runtime comparisons should be made more transparent and apples-to-apples. Overall, the paper makes a valuable contribution with clear insights (post-GRT labels + mixed-size representation) and promising results. I recommend acceptance after addressing the comparative and methodological clarifications noted above.

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